Standard Scaling

Standard scaling normalizes features to have a mean of 0 and a standard deviation of 1: from sklearn.preprocessing import StandardScaler

```
scaler = StandardScaler()
data_scaled = scaler.fit_transform(data)
```

Min-Max Scaling

Min-max scaling transforms features to lie within a specified range, typically [0, 1]: from sklearn.preprocessing import MinMaxScaler

```
scaler = MinMaxScaler()
data_scaled = scaler.fit_transform(data)
```

Feature Selection

Feature selection is the process of identifying the most relevant features for a given task.

By reducing the dimensionality of the dataset, you can:

- Improve computational efficiency.
- Reduce overfitting by eliminating irrelevant or noisy features.
- Enhance model interpretability.

SelectKBest

Scikit-Learn's `SelectKBest` class selects the top `k` features based on a specified statistical test. For classification tasks, the `f_classif` function is commonly used: from sklearn.feature_selection import SelectKBest, f_classif # Select the top 5 features

```
selector = SelectKBest(score_func=f_classif, k=5)
data_selected = selector.fit_transform(data, target)
```

Recursive Feature Elimination (RFE)

RFE iteratively removes the least important features based on model performance: from sklearn.feature_selection import RFE

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
rfe = RFE(model, n_features_to_select=5)
data_selected = rfe.fit_transform(data, target)
```